Samuel Simkin

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Professional Summary

Electrical engineer with automotive electrical-design and system-troubleshooting experience, complemented by independent projects in PCB design, embedded control, current measurement and wireless hardware. Develop electrical documentation and collaborate across engineering teams from design through vehicle builds.

Work Experience

Electrical Design Engineer January 2026 – Present

Avride

- Design new wiring harnesses and revise existing designs for autonomous-vehicle builds based on system requirements and component interfaces.
- Create electrical circuit diagrams, harness drawings and 3D harness models in SOLIDWORKS.
- Develop harness interconnections for sensing, power distribution, communications and vehicle interface systems.
- Coordinate with cross-functional teams from design through final vehicle build, incorporating evolving requirements and design revisions.

Electrical Engineer June 2022 – January 2026

American Honda Motor Company, Marysville, OH

- Performed data acquisition and root-cause analysis using logging and diagnostic tools to troubleshoot vehicle electrical and software integration issues.
- Collaborated with engineering teams on vehicle electrical architecture, signal integrity and interface reliability across CAN, Ethernet, SPI and UART.
- Coordinated with suppliers on part integration, delivery and validation of updates.
- Supported software updates across thousands of vehicles; trained engineers and technicians on update validation procedures and test documentation.

Field Project Supervisor / Engineering Consultant Intern July 2021 – August 2021

Generator Supercenter, Hollywood, FL

- Coordinated project schedules, resources and team performance with branch leadership, resolving bottlenecks during generator installations.
- Oversaw utility-cable rerouting through automatic transfer switch systems for generator installations.

Technical Skills

Design & CAD: Electrical schematics; 3D harness design; PCB layout; Altium Designer; EasyEDA; SOLIDWORKS; AutoCAD; Multisim.

Hardware: Mixed-signal and RF circuits; power management; current measurement; sensor interfaces; MOSFET/relay control; Raspberry Pi; Nordic SoCs.

Test & Programming: Data acquisition; root-cause analysis; digital multimeters; NI myDAQ; Python; LabVIEW; I2C, SPI, UART, CAN and Ethernet.

Selected Hardware Projects

Automotive Battery-Current Measurement System — *PCB design & bench troubleshooting*

- Designed and fabricated an INA260-based current-logging PCB for automotive post-shutdown battery-current measurement, interfacing with a Raspberry Pi.
- Implemented relay-based current-path control using an IRLZ44N MOSFET driver, with a separate USB-C supply for the relay and Raspberry Pi.
- Developed GPIO control logic with software debouncing for switching and data acquisition.
- Compared sensor readings with digital-multimeter measurements and investigated current-path wiring discrepancies and intermittent I2C communication.

Smart Bartender — *Embedded control & software integration*

- Designed Raspberry Pi-based fluid-control electronics using a custom PCB, MOSFET-switched pumps and protected 12 V power circuitry.
- Developed PWM pump-control logic and a Python/Flask interface for user control of dispensing; debugged Flask and SQLite integration on Raspberry Pi.

Bluetooth Audio Device — *PCB & RF design*

- Designed an nRF5340-based Bluetooth audio PCB in Altium Designer, addressing RF layout, audio-signal routing, power management and external amplification.
- Configured Bluetooth profiles for device pairing and LE Audio support, with low-power operation as a design objective.

Bluetooth-Based Tracking Device — *PCB & RF design*

- Designed Nordic-based low-power tracking PCBs using Altium Designer and EasyEDA, including RF matching networks, antenna interfaces and multilayer routing.
- Incorporated crystal oscillators and battery-management circuitry into schematic and design-rule-compliant PCB layouts.

Wearable Sensor Platform — *Component selection & prototype planning*

- Selected nRF52840-DK, BMA400, MAX30101/MAX32664 and TMP117 components for a wearable-sensing concept, with battery monitoring and haptics; planned module-by-module testing ahead of custom-PCB integration.

Research and Projects

Electric Go-Kart August 2021 – May 2022

- Configured a six-battery series/parallel supply for a DC motor and used a digital multimeter to check battery condition and electrical output.

Analog Audio / Karaoke Circuit August 2020 – December 2020

- Designed and built an LF412CN op-amp audio circuit with tone and volume controls, including input, attenuation and output-driver stages.

Education

The Pennsylvania State University May 2022

Bachelor of Science in Electrical Engineering

Languages: Fluent in English and Spanish.

Community Involvement

Consulted on startup projects through TAMID Group at Penn State; supported food fundraising for school workers and tutored elementary students.